

- [20] P. Skrzypczyk, N. Brunner, and S. Popescu, “Emergence of quantum correlations from nonlocality swapping,” *Physical Review Letters*, vol. 102, no. 11, p. 110402, 2009.
- [21] A. J. Short, S. Popescu, and N. Gisin, “Entanglement swapping for generalized nonlocal correlations,” *Physical Review A—Atomic, Molecular, and Optical Physics*, vol. 73, no. 1, p. 012101, 2006.
- [22] J. Barrett, “Information processing in generalized probabilistic theories,” *Physical Review A—Atomic, Molecular, and Optical Physics*, vol. 75, no. 3, p. 032304, 2007.

A Proof of Theorem 3

We prove the second part of the theorem first. Take some arbitrary Φ in $\mathcal{F}_2$. We can represent a Z-channel source as follows

$$P_{XY} = \begin{pmatrix} 1-s & 0 \\ sd & s(1-d) \end{pmatrix}$$

where $s, d \in [0, 1]$. Let $u = f_X(0)$ and $v = f_X(1)$ for some $u, v \geq 0$. Assume that $\mathbb{E}[f] = m = (1-s)u + sv$. Then $u = \frac{m-sv}{1-s}$. One can verify directly that

$$g(v, m) := \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)} = \frac{(1-s(1-d))\Phi\left(\frac{1-s}{1-s(1-d)}u + \frac{sd}{1-s(1-d)}v\right) + s(1-d)\Phi(v) - \Phi((1-s)u + sv)}{(1-s)\Phi(u) + s\Phi(v) - \Phi((1-s)u + sv)} \quad (13)$$

$$= \frac{(1-s(1-d))\Phi\left(\frac{m-s(1-d)v}{1-s(1-d)}\right) + s(1-d)\Phi(v) - \Phi(m)}{(1-s)\Phi\left(\frac{m-sv}{1-s}\right) + s\Phi(v) - \Phi(m)}. \quad (14)$$

Then, we claim that if (10) holds, then $g(v, m)$ is decreasing in v for every fixed m . Therefore, the maximum of $g(v, m)$ would occur when $v = 0$. This would complete the proof. Taking the partial derivative of $\log(g(v, m))$ with respect to v , we need to show that

$$\frac{s\Phi'(v) - s\Phi'\left(\frac{m-sv}{1-s}\right)}{(1-s)\Phi\left(\frac{m-sv}{1-s}\right) + s\Phi(v) - \Phi(m)} \geq \frac{s(1-d)\Phi'(v) - s(1-d)\Phi'\left(\frac{m-s(1-d)v}{1-s(1-d)}\right)}{(1-s(1-d))\Phi\left(\frac{m-s(1-d)v}{1-s(1-d)}\right) + s(1-d)\Phi(v) - \Phi(m)}. \quad (15)$$

For any $t \in [0, 1]$, define

$$k(t) = \frac{st\Phi'(v) - st\Phi'\left(\frac{m-svt}{1-st}\right)}{(1-st)\Phi\left(\frac{m-svt}{1-st}\right) + st\Phi(v) - \Phi(m)} \quad (16)$$

$$= \frac{\Phi'(v) - \Phi'\left(\frac{m-svt}{1-st}\right)}{\left(\frac{1}{st} - 1\right)\Phi\left(\frac{m-svt}{1-st}\right) + \Phi(v) - \frac{1}{st}\Phi(m)}. \quad (17)$$

Then, (15) can be written as $k(1) \geq k(1-d)$. We would be done if we can show that $k(t)$ is an increasing function. Showing $k'(t) \geq 0$ is equivalent with

$$C(t) = -\frac{1}{st^2} \left(\Phi'(v) - \Phi'\left(\frac{m-svt}{1-st}\right) \right) \left(-\Phi\left(\frac{m-svt}{1-st}\right) + \frac{(m-v)st}{1-st} \Phi'\left(\frac{m-svt}{1-st}\right) + \Phi(m) \right) \\ - \frac{s(m-v)}{(1-st)^2} \Phi''\left(\frac{m-svt}{1-st}\right) \left(\Phi(v) + \frac{1-st}{st} \Phi\left(\frac{m-svt}{1-st}\right) - \frac{1}{st} \Phi(m) \right) \geq 0.$$

Let $x_1 = v$, $x_2 = \frac{m-svt}{1-st}$. Then we can compute s from x_1 and x_2 as follows:

$$s = \frac{m - x_2}{t(x_1 - x_2)}.$$